Functional Requirements

Requirement ID	Requirement Description	Category	Priority	Justification
01	The system shall store the following details of the customer: Name, Phone Number, email, Date of birth, Address.	Customer management	Must Have	Essential for customer identification, communication, prescription tracking, and historical records.
02	System shall enable authorised users to add, view, update and deactivate their customers.	Customer management	Must Have	Allows staff to maintain accurate, up-to-date customer files and manage inactive profiles safely.
03	The system shall maintain the details of medicines such as medicine name, medicine generic name, dosage form, medicine strength, medicine price.	Medicine management	Must Have	Critical for accurate dispensing, pricing, and ensuring correct medication information for patients.
04	The system should record the medicine with a unique number.	Medicine management	Must Have	Ensures precise identification and prevents ambiguity or duplication among different medicines.
05	The system shall capture the supplier's information such as name, phone, email and address.	Supplier management	Must Have	Necessary for reordering stock, managing vendor communications, and maintaining supply chain records.
06	The system will be able to link each stock record with a medicine and its supplier.	Inventory management	Must Have	Enables easy traceability of inventory sources and efficient restocking when inventory runs low.
07	The system shall be able to track the number and location of each stock item.	Inventory management	Must Have	Helps staff locate items quickly within the pharmacy and prevents inventory shrinkage or stockouts.
08	A reorder level shall be maintained for each stock item in the system.	Inventory management	Should Have	Automates inventory threshold monitoring to ensure popular or critical medicines never completely run out.
09	The system shall notify authorized personnel if the quantity of the stock is equal to or less than the reorder level.	Inventory management	Should Have	Provides proactive alerts so staff can place purchase orders before stock is entirely depleted.

RQ10	System shall be able to log the expiration dates of medicines and to alert when medicines are running low or have expired.	Inventory management	Must Have	Crucial for patient safety to prevent dispensing expired drugs and to manage waste effectively.
RQ11	The system will log the date, doctor's name, and notes of customer visits.	Visit management	Should Have	Enables tracking of customer consultations, clinical notes, and recommendations over time.
RQ12	The system shall be designed to relate each visit to one customer.	Visit management	Must Have	Ensures visit history and medical consultations are accurately attributed to the correct individual.
RQ13	Prescription information shall be stored such as the date the prescription was issued and the customer the prescription was issued to.	Prescription management	Must Have	Maintains legal compliance and tracks which customer received specific prescribed treatments.
RQ14	There shall be a relation between medicines and prescriptions in the system.	Prescription management	Must Have	Ensures precise fulfillment of exact prescribed drugs and dosages for patients.
RQ15	The system shall generate a sale record of the customer, the date of the sale, the total amount of the sale, and the payment method used for the sale.	Sales management	Must Have	Provides a clear audit trail for financial transactions, reporting, and customer purchases.
RQ16	One or more sale-detail records may be included in a sale.	Sales management	Must Have	Supports transactions where customers purchase multiple distinct items in a single checkout.
RQ17	For each sale-detail record, the system shall log the medicine, quantity, unit price and subtotal.	Sales management	Must Have	Ensures line-item transparency and correct calculation of transaction components.
RQ18	System should decrease the quantity of stock when a sale is made.	Sales management	Must Have	Keeps inventory counts accurate and synchronized immediately after every transaction.
RQ19	The system shall be able to keep a log of	Staff management	Should Have	Aids in accountability, shift reconciliation, and

	who staff processed each sale.			tracking staff performance.
20	The system shall add up the amount of sales from the sale-detail subtotals to provide the sale total.	Sales management	Must Have	Eliminates manual calculation errors during checkout and guarantees financial accuracy.
21	It shall be possible to search for medicines by name.	Medicine management	Should Have	Improves efficiency and speed when staff look up prices or inventory for customers.
22	If the quantity to be sold is greater than the quantity available in the system then the sale shall not be completed.	Sales management	Must Have	Prevents overselling, negative inventory discrepancies, and unfulfilled customer orders.
23	Authorized personnel shall be able to record and update information on suppliers in the system.	Supplier management	Should Have	Ensures contact details and vendor terms remain accurate as supplier relationships evolve.
24	The system shall store a log of transactions for sales that are finished.	Sales management	Should Have	Supports historical reporting, revenue tracking, and auditing requirements.

Non-Functional Requirements

Requirement ID	Requirement Description	Category	Priority	Justification
01	Access to the database for viewing or editing pharmacy information shall be controlled by authentication.	Security	Must Have	Protects sensitive patient health records and financial data from unauthorized access.
02	The database will include primary, foreign and required field constraints that will ensure data integrity.	Data integrity	Must Have	Prevents corrupted data, orphan records, and invalid database states.
03	The database shall be able to finish common searches and lookups within two seconds of the normal operating load.	Performance	Should Have	Ensures fast customer service at the counter without frustrating delays.
04	The database should enable regular backup and recovery of pharmacy records following data loss incidents.	Reliability	Must Have	Protects against accidental deletion, hardware failure, or corruption, ensuring business continuity.
05	The database shall rely on transactions to ensure that any sale is committed with its associated stock changes, or rolled back with its associated changes.	Reliability	Must Have	Prevents inventory mismatches where a sale is recorded without reducing stock, or vice versa.
06	Access to sensitive customer data shall be controlled based on the employees' roles and positions in the database.	Security	Must Have	Enforces the principle of least privilege, safeguarding customer privacy and medical records.
07	The database should be capable of holding at least 50,000 records of medicines, stocks, customers and transactions without having to redesign the schema.	Scalability	Could Have	Accommodates future business growth and historical data accumulation without costly structural re-engineering.

08	The database will maintain dates, prices and quantities in consistent data types and formats.	Maintainability	Must Have	Ensures accurate calculations, prevents sorting/formatting errors, and simplifies reporting queries.